



Assignment 02

Submitted By

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Submitted to

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length() : Returns the length of the string

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

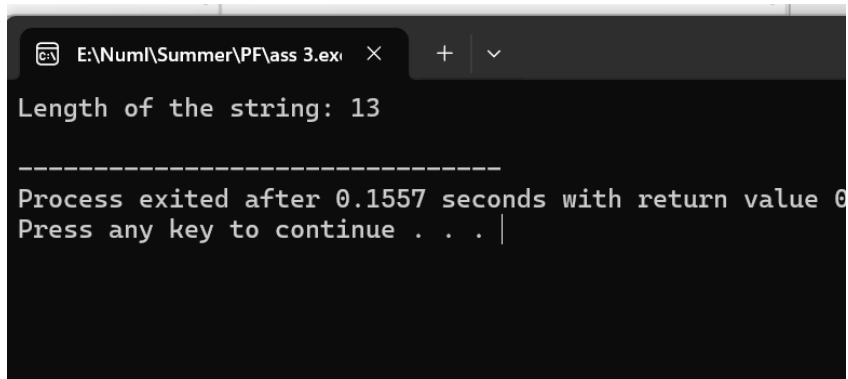
```
int main() {
```

```
    string str = "Hello, World!";
```

```
    cout << "Length of the string: " << str.length() << endl;
```

```
    return 0;
```

```
}
```



```
E:\Num1\Summer\PF\ass 3.exe  X  +  v
Length of the string: 13
-----
Process exited after 0.1557 seconds with return value 0
Press any key to continue . . . |
```

Explanation:

- The `length()` function returns the number of characters in the string `str`.
- For "Hello, World!", the length is 13.

substr() : Returns a substring from a string.

```
#include <iostream>
```

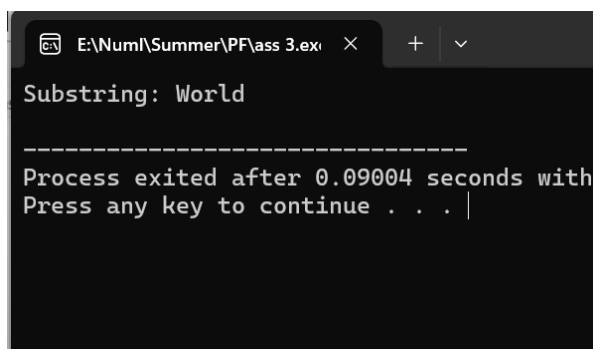
```
#include <string>
```

```
using namespace std;
```

```

int main() {
    string str = "Hello, World!";
    string sub = str.substr(7, 5);
    cout << "Substring: " << sub << endl;
    return 0;
}

```



```

E:\Num1\Summer\PF\ass 3.exe
Substring: World
-----
Process exited after 0.09004 seconds with
Press any key to continue . . .

```

Explanation:

- The `substr(pos, len)` function returns a substring starting at position `pos` and of length `len`.
- Here, it starts at index 7 and returns the next 5 characters, resulting in "World".

find(): Finds the first occurrence of a substring.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string str = "Hello, World!";
```

```

size_t pos = str.find("World");

if (pos != string::npos)

    cout << "'World' found at position: " << pos << endl;

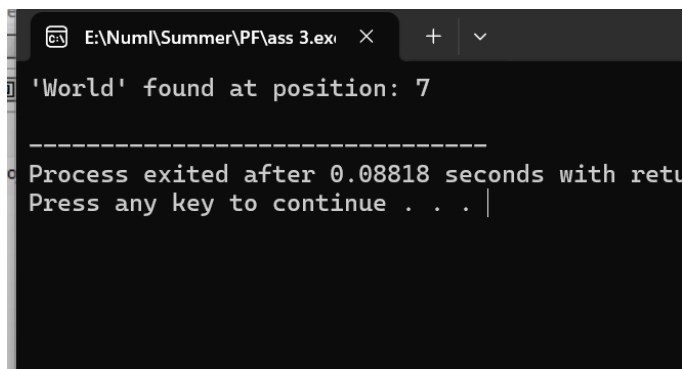
else

    cout << "'World' not found" << endl;

return 0;

}

```



```

E:\Num\Summer\PF\ass 3.exe
'World' found at position: 7
-----
Process exited after 0.08818 seconds with return code 0
Press any key to continue . . .

```

Explanation:

- The `find()` function searches for the first occurrence of the substring "World".
- It returns the position of the first character of the substring.

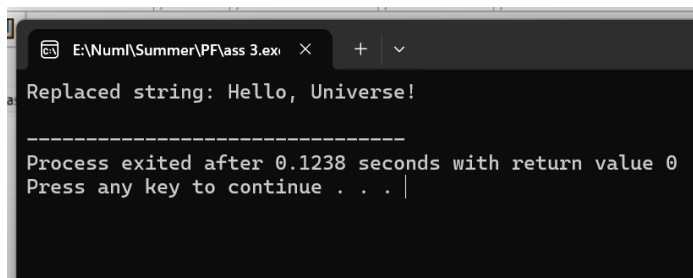
replace () : Replaces part of a string with another string.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {  
  
    string str = "Hello, World!";  
  
    str.replace(7, 5, "Universe");  
  
    cout << "Replaced string: " << str << endl;  
  
    return 0;  
  
}
```



```
E:\Num1\Summer\PF\ass 3.exe  
Replaced string: Hello, Universe!  
-----  
Process exited after 0.1238 seconds with return value 0  
Press any key to continue . . .
```

Explanation:

- The `replace(pos, len, new_str)` function replaces the part of the string starting at `pos` and of length `len` with `new_str`.
- Here, "World" is replaced with "Universe".

append () : Appends a string to another string.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```

int main() {

    string str1 = "Hello";

    string str2 = " World!";

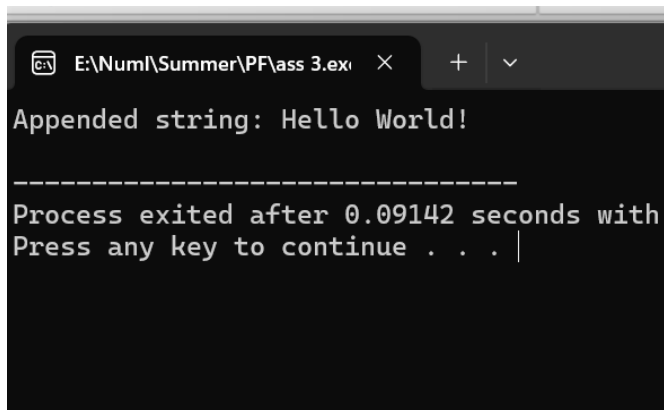
    str1.append(str2);

    cout << "Appended string: " << str1 << endl;

    return 0;

}

```



```

E:\Numl\Summer\PF\ass 3.exe
Appended string: Hello World!
-----
Process exited after 0.09142 seconds with
Press any key to continue . . . |

```

Explanation:

- The `append()` function adds `str2` to the end of `str1`.

erase() : Erases part of a string.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

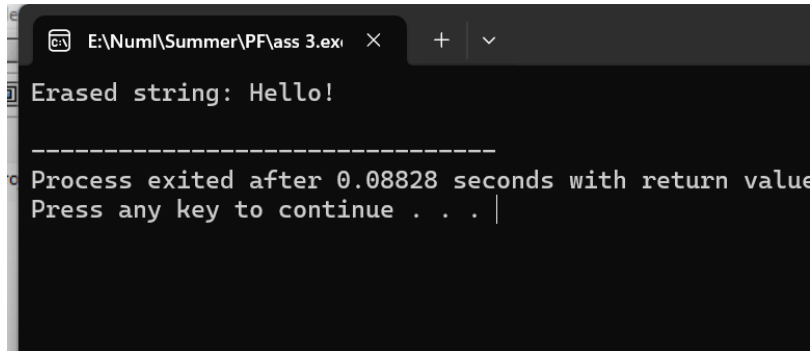
```
    string str = "Hello, World!";
```

```
str.erase(5, 7);

cout << "Erased string: " << str << endl;

return 0;

}
```



```
E:\Num1\Summer\PF\ass 3.exe X + v
Erased string: Hello!
-----
Process exited after 0.08828 seconds with return value 0
Press any key to continue . . . |
```

Explanation:

- The `erase(pos, len)` function removes `len` characters from the string starting at `pos`.
- Here, " World!" is removed.

`insert()`: Inserts a string into another string.

```
#include <iostream>

#include <string>

using namespace std;

int main() {

    string str = "Hello!";

    str.insert(6, " World");

    cout << "Inserted string: " << str << endl;

    return 0;

}
```

```
E:\Num1\Summer\PF\ass 3.exe X + v
Inserted string: Hello, World!
-----
Process exited after 0.1227 seconds with return
Press any key to continue . . . |
```

Explanation:

- The `insert(pos, str)` function inserts `str` at position `pos` in the string.
- " World" is inserted after "Hello, ".

`compare()` : Compares two strings.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string str1 = "Hello";
```

```
    string str2 = "World";
```

```
    int result = str1.compare(str2);
```

```
    if (result == 0)
```

```
        cout << "Strings are equal" << endl;
```

```
    else if (result < 0)
```

```
        cout << "str1 is less than str2" << endl;
```

```
    else
```

```
        cout << "str1 is greater than str2" << endl;
```

```
    return 0;
```

```
}
```



```
E:\NumI\Summer\PF\ass 3.exe X + v
str1 is less than str2
-----
Process exited after 0.09207 seconds w
Press any key to continue . . . |
```

Explanation:

- The `compare()` function compares two strings lexicographically.
- It returns 0 if they are equal, a negative value if `str1` is less than `str2`, and a positive value if `str1` is greater.

`c_str()`: Returns a C-style string (null-terminated).

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

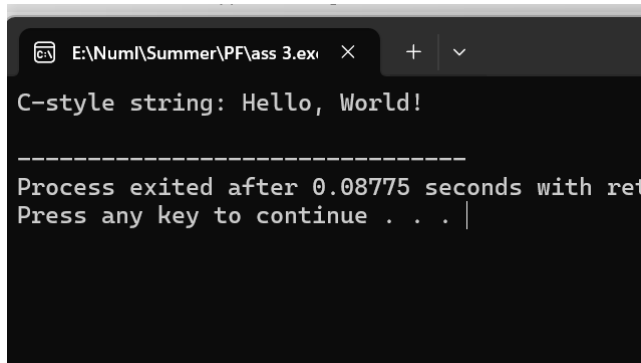
```
    string str = "Hello, World!";
```

```
    const char* cstr = str.c_str();
```

```
    cout << "C-style string: " << cstr << endl;
```

```
    return 0;
```

```
}
```



```
E:\Num\Summer\PF\ass 3.exe
C-style string: Hello, World!
-----
Process exited after 0.08775 seconds with return code 0
Press any key to continue . . . |
```

Explanation:

- The `c_str()` function returns a pointer to a null-terminated C-style string.

`find_first_of()`: Finds the first occurrence of any character from a set.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string str = "Hello, World!";
```

```
    size_t pos = str.find_first_of("aeiou");
```

```
    if (pos != string::npos)
```

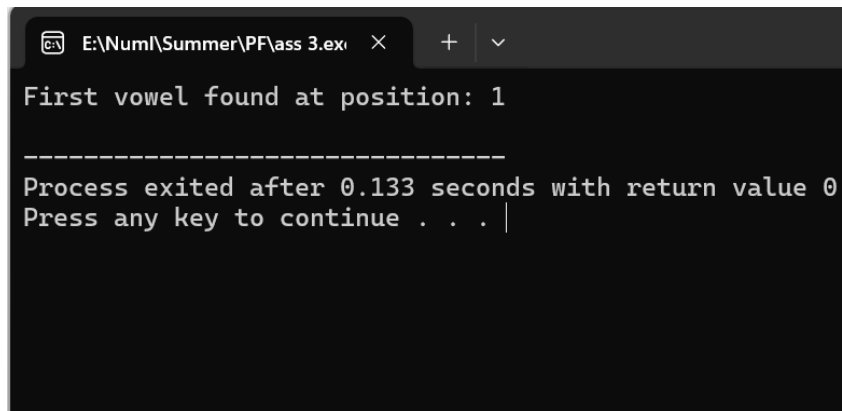
```
        cout << "First vowel found at position: " << pos << endl;
```

```
    else
```

```
        cout << "No vowel found" << endl;
```

```
    return 0;
```

```
}
```



```
E:\Num\Summer\PF\ass 3.exe
First vowel found at position: 1
-----
Process exited after 0.133 seconds with return value 0
Press any key to continue . . . |
```

Explanation:

- This code finds the position of the first vowel (a, e, i, o, u) in the string and prints its position.
- The first vowel in "Hello, World!" is e, found at position 1.